

Week 8

Keel Bone Chicken Project

02/23/26



College of Engineering

This Week

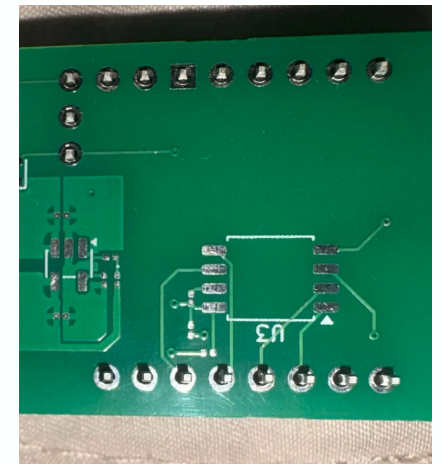
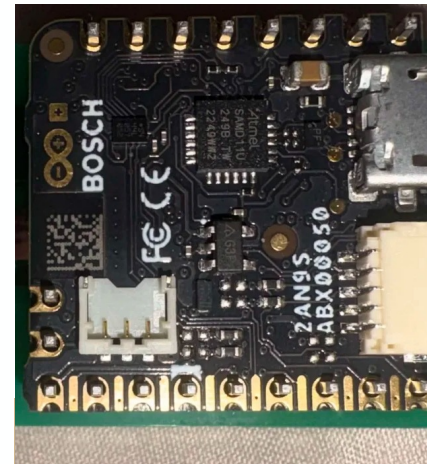
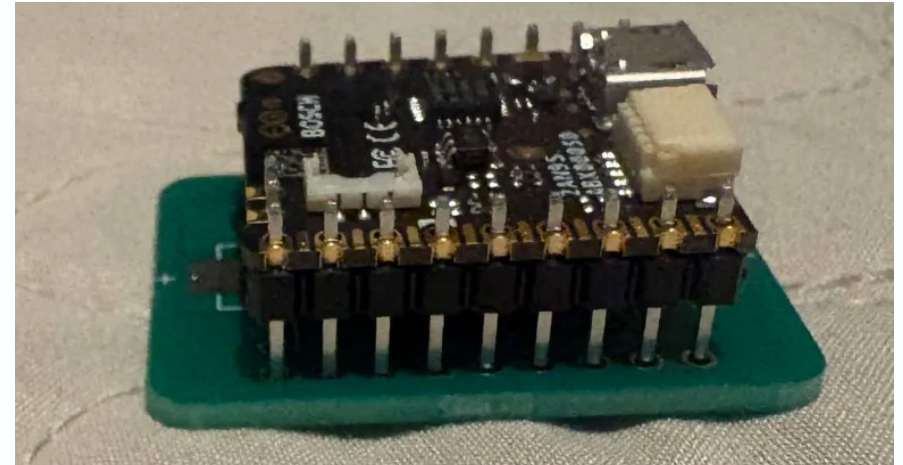
Weekly Progress

- **PCB Updates**
 - Handling of Physical PCBs
 - New Vendor & Vertical Dividers
 - New Layout for Revision 5
 - Placement of RFID Charging on Board
- **Selection of Thermometer**
 - Placement on PCB
- **Purchasing Requests**
 - Sensors
 - Cabling
- **Data Updates & Research**
 - Time Zone Issue Resolved
 - Research into

PCB Updates

Handling of Physical PCBs

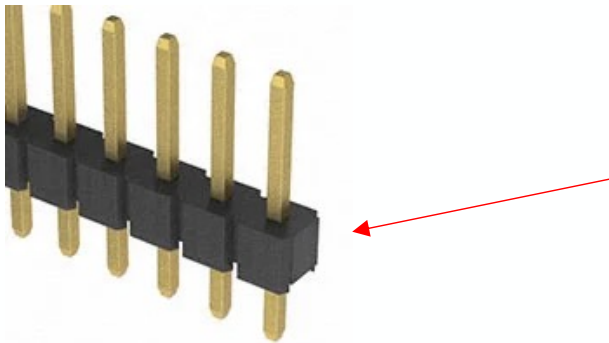
- PCBs have been partially soldered
- Slight alignment errors that need to be analyzed
- Final Approximate Volume:
 - 29.20 mm x 29.20 mm x 11.50 mm = 9.80 cm³



PCB Updates

New Vendor & Vertical Dividers

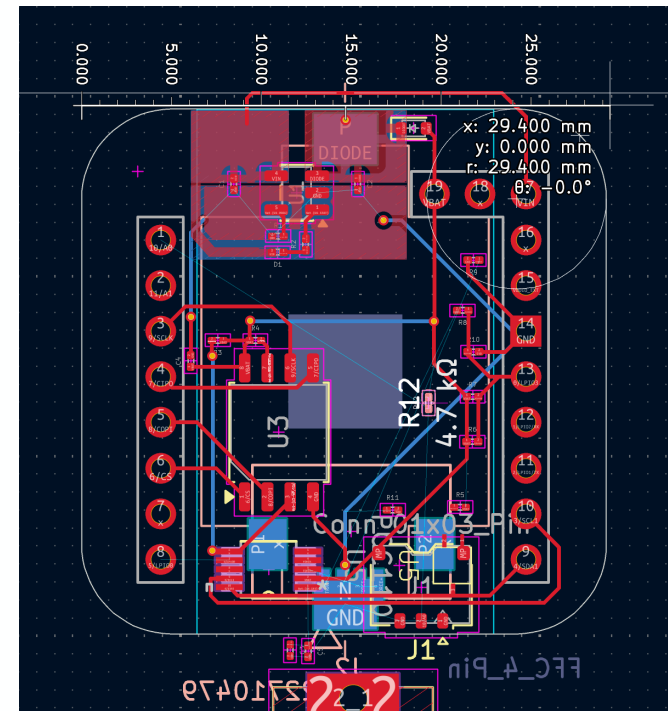
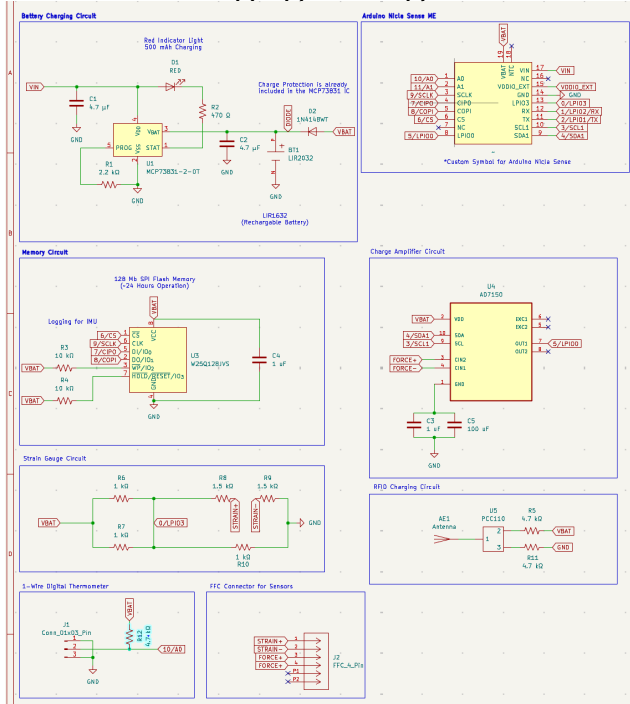
- Black plastic insulating base stripped from existing pin headers and attached to PCB
 - Doubles the distance between the PCBs and allows for better heat dissipation
- Leads to a very tight fit and little heat lost
- New Vendor for PCBs: DL-Engineering, LLC
 - Indiana based company
 - Fast lead times
 - Interaction with engineers for validation
 - Based on Professor Berry's input, have already reached out: <https://dl-engineering.com/contact-us>
 - Also considering 7PCB.com, based on Professor Salas' input: <https://www.7pcb.com>



PCB Updates

New Layout for Revision 5

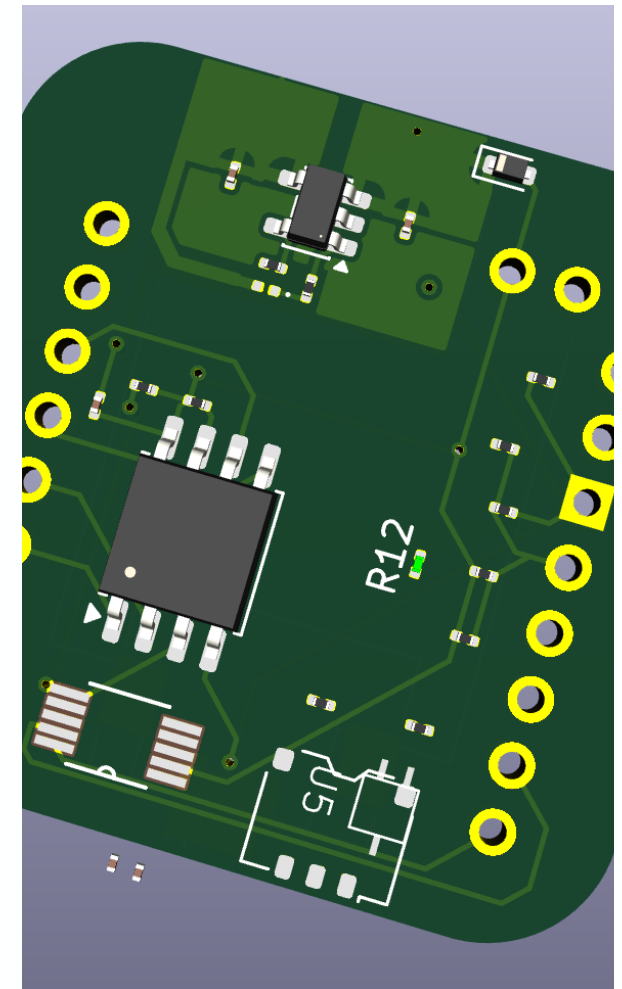
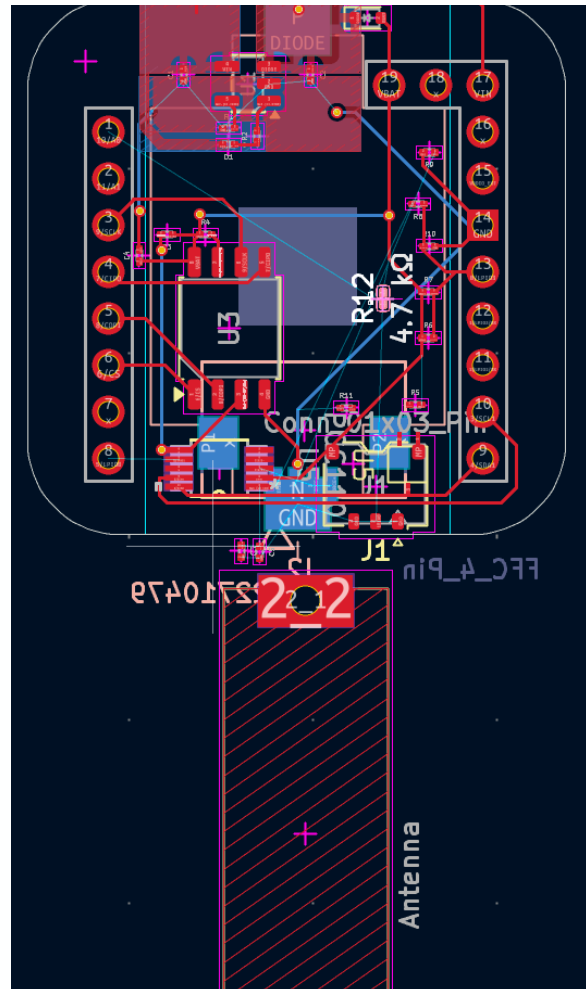
- Size increased to 29.20 mm (~30 mm) square area
 - Sizing expanded to accommodate:
 - 3 Pin JST (Thermistor), FFC Cable, Antenna
 - FFC located with 4 Pins for Charge and Strain
 - RFID Charging Managed with Minimal Circuitry



PCB Updates

Placement of RFID Charging on Board

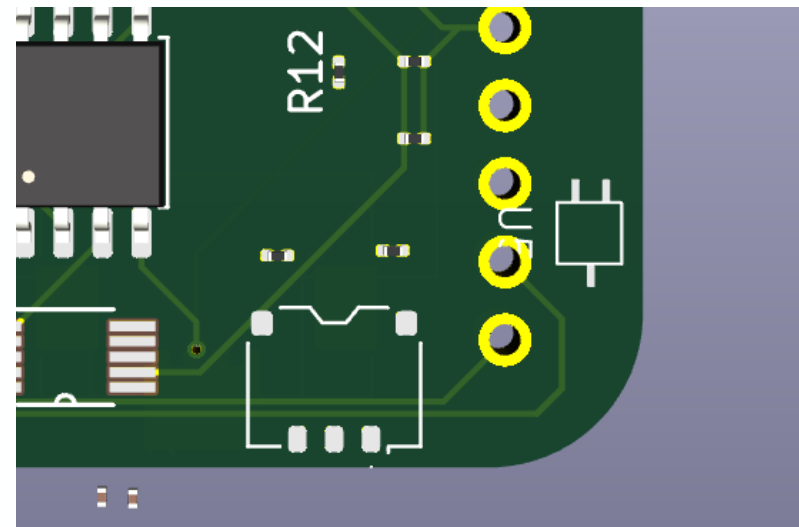
- Positioning of Antenna on Board
- Antenna will hopefully pop OUT of the backpack to receive signal
- ~35 mm antenna (3.5 cm of extrusion)



Selection of Thermometer

Placement on PCB

- Placed at the edge for minimal interference
- Three Pins:
 - Ground
 - Power
 - Signal (LPIO)



Purchasing Requests

Sensors

- Thermometer: SNS-TMP-DS18B20-MAXIM
 - https://www.digikey.com/en/products/detail/olimex-ltd/SNS-TMP-DS18B20-MAXIM/21662296?gclid=CjwKCAiAkVDMBhBMEiwAnUA9BWWfq8euNs4YIOp4R3yMrWfnmcKLHueiwK5gKe5NqGmmtLZWdlv-0xoCUsoQAvD_BwE
- Force: C4A-XX-125SL-120-23P
 - <https://www.digikey.com/en/supplier-centers/micro-measurements>

Purchasing Requests

Cabling

- JST Housings
 - <https://www.adafruit.com/product/261?srltid=AfmBOoqKFq32RkdNJbVjYIHT7c1LhXjM035h4cGkilayPoPf2IxK2BJw>
- DuPont Housings:
 - <https://www.digikey.com/en/products/filter/rectangular-connectors/rectangular-connector-housings/319>
- AWG Wire Housings:
 - <https://www.digikey.com/en/products/detail/te-connectivity-amp-connectors/321026/299225>
- LiPo Battery Chargers:

*JST to DuPont Housings do not exist, the Agriculture department must make them (instruction will be provided)

Next Week

Plans & Blockers

- Obtain data from agriculture and validate data
- Begin first draft of paper

Thank You

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